

Ben Bubnick

HEALTHCARE DATA STRATEGY · DATA GOVERNANCE · ANALYTICS CONSULTING

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Summary

Hands-on engineer specializing in Python on UNIX-like systems (Linux/macOS), production ML/MLOps, and large-scale data pipelines. Heavy daily use of Terminal/Bash/Zsh, SSH, and POSIX tooling to build and operate services. Built forecasting and optimization solutions (e.g., surgery duration prediction, hospitalization risk) and led end-to-end data platforms in AWS and Azure. Strong at turning ambiguous operational problems into real-time services with CI/CD, observability, and documentation. Comfortable collaborating across engineering, data science, and ops—ready to lead AI application development and model integration in high-performance, resource- and cost-sensitive environments.

Experience

HHaEXchange

Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase/GitHub Actions, FHIR/Aidbox

DATA ENGINEERING CONSULTANT — CONTRACT VIA GROWDATA ANALYTICS

Oct. 2025 - Jun. 2026

Led data engineering work spanning source analysis, data modeling, ETL development, identity resolution, and QA to deliver standardized healthcare datasets for reporting and downstream integration.

- Led delivery of Snowflake/dbt pipelines that transformed operational healthcare data into standardized, analytics-ready models.
- Designed organization, office, caregiver, patient, and crosswalk models using FHIR-aligned concepts and stable identity-resolution patterns.
- Built reusable SQL/Jinja transformations for lifecycle fields, null handling, sentinel dates, and source-system variations.
- Developed QA suites covering duplicates, nulls, referential integrity, business rules, and source-to-target reconciliation.
- Partnered with stakeholders to resolve source ambiguity, document mapping standards, and establish reusable onboarding patterns.

Adonis

AWS: S3, Snowflake, SQLMesh, SQL, Python

DATA ENGINEERING CONSULTANT — CONTRACT VIA HEALTH DATA MOVERS

Mar. 2025 - Jul. 2025

Developed a reusable ETL pipeline to streamline data onboarding for a healthcare revenue cycle platform.

- Built a reusable SQLMesh/Snowflake ingestion framework with templated models, data contracts, CI checks, and promotion gates.
- Added schema-drift detection, reconciliation checks, audit logs, and deterministic recovery procedures.
- Developed modular transformations and validation checkpoints that reduced regression risk and accelerated onboarding.
- Translated client requirements into technical specifications, mapping standards, and reusable integration playbooks.

GrowData Analytics

Ruby, RubyLLM, Tesseract OCR, GPT 4o mini, Cladue Opus/Sonnet, Rubocop, RSpec

PRINCIPAL ML ENGINEER

Sep. 2024 - Oct. 2025

Built and shipped an OCR-to-analytics product for small farms; owned model integration, CI/CD, testing, and auditability on UNIX-like systems.

- Built a Python/Ruby OCR service that converted handwritten and printed invoices into structured transaction data.
- Implemented validation rules, audit logs, idempotent reruns, automated testing, and CI/CD for reliable processing.
- Deployed a cost-conscious AWS workflow using S3, Lambda, scheduled jobs, CloudWatch alerts, and operational dashboards.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST

Jun. 2023 - Jan. 2025

Built a scalable AWS data lake with automated CI/CD, metadata reporting, and integrated analytics tools.

- Built an AWS data lake with dbt-based CI/CD, cost-aware partitioning, and job scheduling, cutting deployment time by approximately 50%.
- Improved reliability through pipeline monitoring, failure classification, alerting, and incident-response runbooks.
- Integrated Jupyter-based reporting and documented the platform, improving analytical usability and developer onboarding.
- Mentored engineers and established standards for code review, testing, pipeline design, and stakeholder delivery.

Merative

Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST

Jul. 2022 - Mar. 2023

Launched a new Azure platform to replace HDFS systems to modernize data infrastructure.

- Led HIPAA-compliant deidentification across more than 200 million patient records.
- Built automated regression and data-comparison tests to detect schema drift and transformation errors.
- Validated migration workflows across Azure Data Factory, Blob Storage, SQL Database, and Synapse Analytics.
- Documented migration procedures and enforced governance controls supporting HITRUST readiness.

IBM

Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST

Nov. 2021 - Jul. 2022

Built AI tools while leading analytics delivery and large-scale data validation efforts.

- Built surgical scheduling optimization and CNN/RNN case-card solutions.
- Led client analytics delivery and validated the migration of more than 150 billion patient records.

SENIOR DATA ARCHITECT

Dec. 2020 - Nov. 2021

Led large-scale clinical data and NLP initiatives and delivering ML solutions.

- Led integration of more than 100 billion clinical records and topic modeling across nearly six million provider notes.
- Built a COVID-19 hospitalization model while documenting AI workflows and mentoring new hires.

DATA ARCHITECT

Nov. 2017 - Dec. 2020

Led enterprise data integration initiatives and mentoring programs.

- Managed multi-org enterprise integration with over 70 interconnected data sources.
- Developed new data validation and review process that reduced total rework by 25%.

SOFTWARE DEVELOPER, DATA PLATFORM

Sep. 2016 - Nov. 2017

Developed internal resources that reduced costs and improved onboarding efficiency.

- Built a data-curation tool that replaced a \$30,000 annual contract.
- Redesigned documentation and training, reducing knowledge-transfer time by 80%.

DATA SCIENTIST

Nov. 2015 - Sep. 2016

Led integration efforts and process automation initiatives.

- Managed 12 remote contractors across 36 data-integration projects.
- Built automation that eliminated 28% of unnecessary development work.

Explorys

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENCE CONSULTANT — CONTRACT VIA TEKSYSTEMS

Jun. 2015 - Nov. 2015

Built ETL pipelines on distributed data systems to support clinical data integration and population analytics.

- Served as principal integration specialist for more than 10 client data warehouses.
- Troubleshoot production pipelines and built scheduling controls for client data ingestion.

AmTrust Financial Services

VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

SOFTWARE ENGINEER

Mar. 2014 - Jun. 2015

Developed MVC components and web forms for insurance policy software.

- Led development of insurance policy software and a new product offering.
- Analyzed 65+ rating forms to identify and resolve performance bottlenecks.
- Introduced unit testing for multi-tier MVC components.

PPG Industries

VB.NET, C#, SQL, C++, FORTRAN

COLOR SCIENTIST

Apr. 2012 - Jun. 2013

Developed numerical methods and software for automotive color matching and measurement.

- Improved color-matching classification accuracy by 40% through advanced numerical methods.
- Built cross-instrument correlation software that improved measurement consistency and QA reliability.
- Developed formulation and pigment-analysis methods for an automated automotive color-matching facility.

Multiple Institutions

ADJUNCT PROFESSOR

Aug. 2010 - May. 2019

Taught physics & astronomy lecture series and lab courses.

- Taught physics and astronomy courses using interactive, peer-led, and just-in-time instructional methods.
- Produced more than 700 hours of recorded lectures and lab material for on-demand learning.
- Exceeded evaluation benchmarks by 16–17% year over year.

Professional Recognition

HDM, SPECIAL RECOGNITION (2025), Recognized for exceptional performance and commitment in client delivery.

Arkos, ARKOS HEALTH BI STAR (2023), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (2022), Recognized for systems migration efforts.

IBM, GOLD CHAMPION (2018 - 2022), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

IBM, LAB SERVICES AWARD (2018), Recognized for coordinating internal teams to deliver solutions on time.

IBM, EMINENCE AND EXCELLENCE AWARD (2017), Recognized for developing a new data curation tool.

PPG, COLOR LAB AUTOMATION AWARD (2012), Recognized for expanding data collection and analysis reporting.

Volunteer

HSEA

ETL BRIGADE

Feb. 2025 - May. 2025

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll

INSTRUCTOR

Oct. 2018 - May. 2019

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE

COACH

Mar. 2017 - Mar. 2019

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America

ETL GUILD

Jan. 2015 - Jun. 2015

The initiative was brought about to apply data science tools to improve and expand public record access.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization

"An algorithm uses past utilization and current waitlist info to predict surgery duration.

Dec. 2020

IBM

REACH Initiative: IBM Watson Care Insights

Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.

2019

IBM

Content & Data Analytics Learning Exchange

Introduction of implementation of machine learning algorithms using real-world health data analytics.

2018

IBM

2018 Innovation Research Interchange Fall Networks Conference

Using NLP to extract insight from unstructured data and transform it from a data lake.

Sep. 2018

IRI

Cross Team Training Series

A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.

2016

EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets

Reproducible Postgres load of Tuva seed datasets with robust validation.

Aug. 2025

HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction ^a

Improvement on traditional fixed ratio methods for total procedure time estimation.

Dec. 2020

IBM

Process and Requirements for CCD Data Integration

Processes and requirements for overcoming the many difficulties with CCD data curation.

Oct. 2017

IBM

AI Paint Formula Prediction from Spectroscopic Measurements ^b

Naive bayesian color and effects matching algorithm of complex automotive paint formula.

May. 2015

SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching

Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.

Feb. 2013

USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy ^c

Spectroscopic and Photometric analysis of two open clusters in the galactic plane.

Dec. 2010

OHIOLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107

Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.

May. 2008

VIZIER

Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107

New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).

Feb. 2008

PASP

A Novel Method for Low-Toxic Photo-Etch Printing ^d

Application of aquatint etching techniques to photo-etch procedures.

May. 2003

MAPC

Education

2022 **PGCert.**, Data Science Professional^a

IBM

2015 **PGCert.**, Data Science^b

Johns Hopkins University

2010 **Master of Science**, Astrophysics^c

University of Cincinnati

2004 **Bachelor of Fine Arts**, Printmaking^d

Ohio University